United States Patent	12403797
Kind Code	B2
Date of Patent	September 02, 2025
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Seat device

Abstract

A seat device (**10**) that can reliably and easily regulate a reclining operation with a simple and compact configuration, only in a specific state where the reclining operation of a seat (**1**) is problematic. It includes a reclining operation unit (**100**) that performs an operation of tilting a backrest (**3**) of the seat (**1**) by a reclining mechanism (**50**), a lock mechanism (**120**) that restrains the reclining operation unit (**100**) to disable the operation when the seat (**1**) is in a long state, and a cable (**124**) that is pulled in conjunction with conversion into the long state of the seat (**1**). The lock mechanism (**120**) includes a stopper (**121**) that is linearly moved to be engaged with and released from an operation lever (**110**) of the reclining operation unit (**100**), and the stopper (**121**) is directly pulled by the cable (**124**) to be linearly moved to a restrained position at which the stopper (**121**) is engaged with the operation lever (**110**) to disable the operation.

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Appl. No.:	**18/254873**
Filed (or PCT Filed):	**November 10, 2021**
PCT No.:	**PCT/JP2021/041297**
PCT Pub. No.:	**WO2022/113737**
PCT Pub. Date:	June 02, 2022

Prior Publication Data

Document Identifier	**Publication Date**

Foreign Application Priority Data

JP	2020-198656	Nov. 30, 2020

Publication Classification

Int. Cl.: **B60N2/14** (20060101); **B60N2/06** (20060101); **B60N2/20** (20060101)

U.S. Cl.:

CPC **B60N2/14** (20130101); **B60N2/06** (20130101); **B60N2/20** (20130101);

Field of Classification Search

CPC: A47C (1/024); A61G (5/1067); B61D (33/021); B60N (2/14); B60N (2/06); B60N (2/20)

USPC: 297/354.12

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Background/Summary

TECHNICAL FIELD
(1) The present invention relates to a seat device that can convert the state of a seat.
BACKGROUND ART
(2) Conventionally, there are seats of a stool type mounted in, for example, railroad cars, which are long in both directions and can seat a plurality of persons, and are generally installed along walls in cabins. Among the stool type seats, a rotary seat is known that can be rotated about a rotation axis in the center of the seat to convert the orientation of the seat between a long state where the back of the seat is parallel to and along a wall, and a cross state where the back of the seat is orthogonal to the wall.
(3) As for such a rotary seat, a seat device has been proposed that includes a sliding mechanism in addition to a rotation mechanism of the seat, and further includes a transmission mechanism for interlocking each mechanism, so that the trajectory (turning radius) of the seat does not interfere with a wall, when rotating the seat from the long state along the wall to the cross state (refer to, for example, Patent Literature 1).
(4) In the seat device, in order to expand the aisle width between seats on both sides in a cabin as much as possible to obtain a comfortable space, the rotation axis of the seat is located near a wall in the long state, while the rotation axis of the seat is slid to an aisle side in the cross state, so that the seat does not interfere with the wall. Therefore, in the seat device, in order to prevent the interference with the wall in the long state, a backrest could not be tilted, and a reclining mechanism for improving seating comfort could not be provided.
(5) Therefore, the present inventors have proposed, in Japanese Patent Application No. 2020-34280, a seat device that regulates a reclining operation when a seat is in the long state. This seat device disables the reclining operation by a lock mechanism provided in an armrest, when the seat is in the long state. The lock mechanism restrains a reclining operation lever by using an L-shaped link and a lock pin that are interlocked for conversion of the state of the seat. The L-shaped link is a member that is long in the vertical direction, and the lock pin is operated in the vertical direction by the L-shaped link.
(6) Additionally, as another rotary seat, a rotary seat is proposed that includes an anti-movement mechanism for disabling the reclining mechanism when the seat faces a window side (for example, see Patent Literature 2).
(7) The anti-movement mechanism is provided with long links on both sides of a bottom surface side (movable base) of the seat, a reclining operation wire is connected to one end side of the links, a wire of the reclining operation lever as well as a tension spring are connected to the other end side, and when the seat faces the window side, the tension spring is pulled, and the wire of the operation lever is disabled to be towed.
PATENT LITERATURE
(8) Patent Literature 1: Japanese Patent No. 3431772 Patent Literature 2: Japanese Utility Model Publication No. 63-13947

SUMMARY OF THE INVENTION

(9) However, in the conventional technique shown in Japanese Patent Application No. 2020-34280 described above, since the lock mechanism provided in the armrest with the reclining operation lever restrained the operation lever by using the L-shaped link and the lock pin, there were large number of parts, the configuration was complicated, and there was a risk of causing a high cost. Additionally, since the L-shaped link was long and bulky in the vertical direction, a substantial arrangement space, which also included its range of movement, was also increased, a limited space for the armrest was eroded, and there was a design problem that it was limited to add, for example, a cushioned elbow pad.

(10) In addition, also in the conventional technique described in Patent Literature 2, since the anti-movement mechanism restrains the operation lever by using a link and a tension spring, there were large number of parts, the configuration was complicated, and there was a risk of causing a high cost. Further, since the links were long and bulky on both sides, a larger arrangement space, which also included its range of movement, was required, and it was difficult to provide the links in the bottom surface side (movable base) of the seat, which is a particularly limited space.

(11) Moreover, the aforementioned anti-movement mechanism did not directly restrain the movement of the operation lever, but indirectly restrained the movement of the operation lever via the wire, the links, and the tension spring. Accordingly, there was a problem that, for example, when the tension spring is deteriorated, restraining of the operation lever became insufficient, and locking lacks certainty.

(12) The present invention has been made by focusing on the problems of the related art as described above, and an object of the present invention is to provide a seat device that can reduce the cost with a simple configuration having a reduced number of parts, can respond to a request of space-saving by enabling a compact configuration, and can reliably and easily regulate a reclining operation only in a specific state where the reclining operation is problematic.

(13) In order to achieve the aforementioned object, one aspect of the present invention is a seat device that can convert a state of a seat, including: a reclining mechanism that can tilt a backrest of the seat; an operation unit that performs an operation of tilting the backrest by the reclining mechanism; a lock mechanism that restrains the operation unit to disable the operation when the seat is in a specific state; and a cable that is pulled in conjunction with conversion into the specific state of the seat, wherein the operation unit can be displaced from an ordinary initial position to an operation position at a time of the operation, the lock mechanism includes a stopper that is linearly moved to be engaged with and released from the operation unit in the initial position, and the stopper is directly pulled by the cable to be linearly moved from an ordinary unrestrained position at which the stopper is released from the operation unit to a restrained position at which the stopper is engaged with the operation unit to disable the operation.

(14) With the seat device according to the present invention, it is possible to reduce the cost with a simple configuration having a reduced number of parts, to respond to a request of space-saving by enabling a compact configuration, and to reliably and easily regulate a reclining operation only in a specific state where the reclining operation is problematic.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a perspective view showing a reclining operation unit of a seat device according to an embodiment of the present invention.

(2) FIG. **2** is a side view showing the reclining operation unit of the seat device according to the embodiment of the present invention.

(3) FIG. **3** is a perspective view showing the reclining operation unit of the seat device according to

the embodiment of the present invention, with a part of the reclining operation unit being omitted.
(4) FIG. **4** is a perspective view and a side view showing the reclining operation unit of the seat device according to the embodiment of the present invention.
(5) FIG. **5** is a perspective view showing a cross state of the seat device according to the embodiment of the present invention.
(6) FIG. **6** is a perspective view showing a long state of the seat device according to the embodiment of the present invention.
(7) FIG. **7** is a perspective view showing an internal structure of the seat device according to the embodiment of the present invention.
(8) FIG. **8** is a front view showing the cross state of the seat device according to the embodiment of the present invention.
(9) FIG. **9** is a side view showing the long state of the seat device according to the embodiment of the present invention.
(10) FIG. **10** is a perspective view showing a leg stand, a movable stand, and an underframe of the seat device according to the embodiment of the present invention.
(11) FIG. **11** is a side view showing a rotation operation unit of the seat device according to the embodiment of the present invention.
(12) FIG. **12** is a side view showing the rotation operation unit of the seat device according to the embodiment of the present invention.
(13) FIG. **13** is a perspective view showing a rotation lock mechanism of the seat device according to the embodiment of the present invention.
(14) FIG. **14** is a perspective view showing the rotation lock mechanism of the seat device according to the embodiment of the present invention.
(15) FIG. **15** is an explanatory diagram showing processes of converting a state of the seat in the seat device according to the embodiment of the present invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(16) Hereinafter, an embodiment representing the present invention will be described based on the drawings.
(17) FIG. **1** to FIG. **15** show one embodiment of the present invention.
(18) A seat device **10** according to the present embodiment can convert the state of a seat **1**. Here, the state of the seat **1** is a concept including not only the orientation of the seat **1** by rotation, but also the change in the front and back position of the seat **1**, etc. Additionally, although the kind of the seat **1** is not particularly limited, a case will be described below as an example where the seat **1** is applied to a stool for two persons mounted in a cabin of a railroad car. Note that, in each drawing, the relative size relationships, shapes, and the like of components may be appropriately designed and changed, and may be different from actual relative size relationships, shapes, and the like of components.
(19) <Outline of Seat Device **10**>
(20) FIG. **10** is a perspective view showing a leg stand **11**, a movable stand **20**, and an underframe **30** of the seat device **10**. FIG. **7** is a perspective view showing an internal structure of the seat device **10**. FIG. **5** is a perspective view showing a cross state of the seat device **10**, and FIG. **8** is a front view also showing the cross state of the seat device **10**. FIG. **6** is a perspective view showing a long state of the seat device **10**, and FIG. **9** is a side view also showing the long state of the seat device **10**. Note that, in each diagram, slight differences in the shapes of identical parts are merely design changes (for example, a difference in the shape of the leg stand **11** in FIG. **5** and FIG. **10**, and the like).
(21) As shown in FIG. **10**, the seat device **10** includes a leg stand **11** fixed on a floor surface, a movable stand **20** supported by the leg stand **11** so as to be able to advance and retract in front and back directions, and the underframe **30** of the seat **1** supported by the movable stand **20** so as to be rotatable in forward and backward directions. Here, the leg stand **11** is an example of “a fixed side

of the seat” of the present invention, and the movable stand **20** and the underframe **30** are examples of “a movable side of the seat” of the present invention. Note that the seat device **10** is arranged on the floor surface near a wall (window) in the cabin of the railroad car, and “A” in FIG. **10** is a part of the wall parallel to the moving direction of the railroad car.

(22) In the seat device **10**, the underframe **30** of the seat **1** is supported on the movable stand **20** to be rotatable about a rotation axis via the rotation mechanism **40**. Additionally, the movable stand **20** is supported on the leg stand **11** together with the rotation mechanism **40** via a sliding mechanism **14** so as to be able to advance and retract. Furthermore, although an illustration is omitted, the seat device **10** includes an interlocking mechanism in order to interlock the rotation of the seat **1** by the rotation mechanism **40** with the advancement and retraction of the seat **1** by the sliding mechanism **14**.

(23) <Regarding Seat **1**>

(24) As shown in FIG. **5**, the seat **1** is formed as a stool for two persons by arranging a pair of seating portions **2** and backrests **3** side by side in both directions. A pair of sleeve portions **4** covering the seating portions **2** from the sides are provided on both sides of the seat **1**. An upper end side of each sleeve portion **4** serves as an armrest **5** that extends substantially horizontally in front and back directions. Here, the armrest **5** is configured by fitting a cushioned elbow pad on an upper end surface of a frame member, which forms a sleeve portion **4**. Additionally, a lower end side of the backrest **3** is supported at a rear end side of the seating portion **2** via a reclining mechanism **50** (refer to FIG. **7**) in a tiltable manner.

(25) <Reclining Mechanism **50**>

(26) As shown in FIG. **7**, the reclining mechanism **50** supports the backrest **3** with respect to the seating portion **2** in a predetermined angular range in a tiltable manner. The reclining mechanism **50** includes, for example, a damper **51** such as a gas spring. The damper **51** is configured such that a piston rod **53** is inserted into a cylinder body **52** so as to be able to protrude, and is arranged on the underframe **30** to be parallel to the front and back directions. Although the damper **51** is urged in a direction in which the piston rod **53** is housed in the cylinder body **52**, the piston rod **53** can be fixed in a state where only an arbitrary amount of the piston rod **53** is projected.

(27) A rear end of the cylinder body **52** is connected to a lower end of a frame **3***a* of backrest **3** so as to be able to be pushed and pulled. On the other hand, a tip of the piston rod **53** that protrudes from a front end of the cylinder body **52** is connected to a proper place of a front end side of the underframe **30**. With such a damper **51**, the backrest **3** can be held at an arbitrary tilt angle. That is, when the damper **51** is a locked state, the piston rod **53** is fixed in a state where only a predetermined amount of the piston rod **53** is projected from the cylinder body **52**, so that the backrest **3** can be held at an arbitrary tilt angle.

(28) When the locked state of the damper **51** is released, the backrest **3** returns to a most upright initial position by the biasing force with which the piston rod **53** is housed in the cylinder body **52**. When a seated person presses the backrest **3** backward against the restoring force of the damper **51** in this lock released state, the backrest **3** can be adjusted to an arbitrary tilt angle.

(29) Although a detailed description of the lock mechanism of the damper **51** is omitted since the lock mechanism is common, a release button **54** for releasing the locked state is provided near the piston rod **53**. As shown in FIG. **7**, a reclining operation unit **100** for pressing this release button **54** to release the locked state is provided in a front end of the armrest **5**. Note that the reclining operation unit **100** is an example of the “operation unit” of the present invention.

(30) <Reclining Operation Unit **100**>

(31) FIG. **1** is a perspective view of the reclining operation unit **100**, FIG. **1**(*a*) shows a state where the operation lever **110** is not restrained (hereinafter “the unrestrained state”), and FIG. **1**(*b*) shows a state where the operation lever **110** is restrained (hereinafter “the restrained state”). FIG. **2** is a side view of the reclining operation unit **100**, FIG. **2**(*a*) shows the unrestrained state of the operation lever **110**, and FIG. **2**(*b*) shows the restrained state of the operation lever **110**.

(32) FIG. **3** is a perspective view of the reclining operation units **100**, with a part (flange **105**) of the reclining operation unit **100** being omitted, FIG. **3**(*a*) shows the unrestrained state of the operation lever **110**, and FIG. **3**(*b*) shows the restrained state of the operation lever **110**. FIG. **4** shows an operation position at which the operation lever **110** of the reclining operation unit **100** is lifted, FIG. **4**(*a*) is a perspective view, and FIG. **4**(*b*) is a side view. Note that, as shown in FIG. **1** to FIG. **4**, the reclining operation unit **100** is provided with a reclining operation lock mechanism **120**.
(33) As shown in FIG. **1** to FIG. **4**, the reclining operation unit **100** is unitized by incorporating each part in one housing **101**, and is incorporated inside a front end of the armrest **5** (to be precise, an upper end side of the sleeve portion **4**) as shown in FIG. **5**. The reclining operation unit **100** performs an operation (reclining operation) for tilting the backrest **3** by the reclining mechanism **50**. The housing **101** is a metal member having a long shape in front and back directions as shown, and a bottom wall **103** is bent at a right angle along a lower end edge of a side wall **102**.
(34) The reclining operation unit **100** includes the operation lever **110** swingably supported by the housing **101**. The side wall **102** of the housing **101** is provided with a supporting wall **104** that is parallel to the side wall surface **102**. The operation lever **110** is swingably supported at a substantially center of the entire housing **101**, which is a rear end side of the supporting wall **104**, via an axis **111** extending in horizontal directions on a horizontal plane.
(35) The operation lever **110** is formed into a substantially L-shape, and one end side serves as an operation unit **112** that is operated by a seated person with fingers, with the axis **111**, at which the substantially middle of the operation lever **110** is pivotably supported, serving as an oscillation center. The operation unit **112** is arranged toward the front of the seat **1**, and is swung mainly in the vertical direction. The other end side, which is located on the other side of the oscillation center with respect to the operation unit **112**, is provided with an abutting portion **113** that is engaged with and released from a stopper **121**, which will be described later. The abutting portion **113** is arranged in a downward direction, and is swung mainly in the front and back directions. A concave groove **114** is provided in a lower portion of the abutting portion **113**, and one end side of a reclining lock cable **124**, which will be described later, is inserted into the concave groove **114**.
(36) The operation lever **110** is set to oscillate (displace) from an ordinary initial position at which the operation unit **112** faces downward and the abutting portion **113** becomes closer to the rear as shown in FIG. **2**(*a*) to an operation position at which the operation unit **112** is pulled upward and the abutting portion **113** becomes closer to the front at the time of the reclining operation as shown in FIG. **4**(*b*). Additionally, a hanging groove **115** to which one end side of a reclining operation cable **116** is connected is provided between the oscillation center (axis **111**) and the other end side (abutting portion **113**) of the operation lever **110**.
(37) The other end side of the reclining operation cable **116** extends to a release button **54** side (see FIG. **7**) of the reclining mechanism **50**, although an illustration is omitted. Here, by pulling the operation lever **110** upward to the operation position, a link (not shown) on the release button **54** side is pulled via the reclining operation cable **116** and the release button **54** is pressed, so that the locked state of the damper **51** is released. Note that the operation lever **110** is normally pulled by the reclining operation cable **116** to be held at the initial position.
(38) <Reclining Operation Lock Mechanism **120**>
(39) The reclining operation unit **100** is provided with the reclining operation lock mechanism **120** that can restrain the reclining operation unit **100** to disable the operation. The reclining operation lock mechanism **120** restrains the reclining operation unit **100** to disable the operation, when the seat **1** is in the long state (specific state), which will be described later. Note that the reclining operation lock mechanism **120** is an example of “a lock mechanism” of the present invention.
(40) As shown in FIG. **1** to FIG. **4**, the reclining operation lock mechanism **120** includes the stopper **121** that is engaged with and released from the abutting portion **113** of the operation lever **110**, when the operation lever **110** of the reclining operation unit **100** is in the initial position. The stopper **121** is supported on the bottom wall **103** of the housing **101** at a position closer to the front,

so as to linearly move in the front and back directions of the seat **1**.
(41) The stopper **121** is formed from, for example, a metal spinning-top-shaped member, and is supported at a position closer to the front of the bottom wall **103**, so as to linearly move in the front and back directions on the bottom wall **103**, which is a substantially horizontal plane, along the flange **105** that rises parallel to the side wall **102**. Here, the flange **105** is provided with a guide groove **106** to which an axis **122** protruding from a side of the stopper **121** movably fits, and by which the axis **122** is guided. Note that the stopper **121** is located between the operation unit **112** and the abutting portion **113** of the operation lever **110** in plan view.
(42) The one end side of the reclining lock cable **124**, which is pulled when the seat **1** is converted into the long state (specific state) described later, is connected to the stopper **121** from behind. The one end side of the reclining lock cable **124** is linearly routed to extend in the same direction as the direction in which the stopper **121** is linearly moved. Here, the one end side of the reclining lock cable **124** extends forward while being inserted into the concave groove **114** in the abutting portion **113** of the operation lever **110**, and is directly connected to a backside of the stopper **121** located ahead of the abutting portion **113**.
(43) On the other hand, the other end side of the reclining lock cable **124** extends to a rotation operation unit **200** side, which will be described later. The reclining lock cable **124** is set to be pulled from the rotation operation unit **200** side, when the seat **1** is in the specific state (the long state described later). When the stopper **121** is directly pulled by the reclining lock cable **124**, the stopper **121** is linearly moved from an ordinary unrestrained position (see FIG. **1**(*a*)) at which the stopper **121** is released from the abutting portion **113** of the operation lever **110** to a restrained position (see FIG. **1**(*b*)) at which the stopper **121** is engaged with the abutting portion **113**. Note that the reclining lock cable **124** is an example of “a cable” of the present invention.
(44) A spring member **123** is stretched between a front side of the stopper **121** and a front end piece **107** of the housing **101** located ahead of the stopper **121**. The spring member **123** urges the stopper **121** to the ordinary unrestrained position. As shown in FIG. **1**(*a*), when the stopper **121** is at the unrestrained position, i.e., a forefront end of a linear moving range, the swing of the operation lever **110** is not restrained, and as shown in FIG. **4**, the reclining operation is enabled that causes the operation unit **112** of the operation lever **110** to swing to an upper operation position. Note that the spring member **123** is an example of “biasing means” of the present invention.
(45) When the reclining lock cable **124** is pulled, the stopper **121** is linearly moved backward to the restrained position, i.e., a rear end of the linear moving range, against the biasing force of the spring member **123** as shown in FIG. **1**(*b*). Accordingly, since a rear end surface of the stopper **121** is engaged with the abutting portion **113** of the operation lever **110**, the operation lever **110** remains at the ordinary initial position, cannot swing the operation unit **112** upward, and is restrained to disable the operation. Note that the supporting wall **104** of the housing **101** serves to define the initial position of the operation lever **110**.
(46) Additionally, the reclining lock cable **124** is formed by slidably inserting an inner cable into an outer cable, an end of the outer cable is fixed to a bracket **108** provided on a rear end side of the side wall **102** of the housing **101**, and the inner cable, which forms the one end side of the reclining lock cable **124**, extends to the stopper **121** ahead of the bracket **108**. In addition, similarly, the reclining operation cable **116** is also formed by slidably inserting an inner cable into an outer cable, an end of an outer cable is fixed to the bracket **108**, and the inner cable, which forms the one end side of the reclining operation cable **116**, extends to the hanging groove **115** of the operation lever **110**.
(47) <Regarding State of Seat **1**>
(48) FIG. **15** is an explanatory diagram showing the process of converting the seat **1** into the long state, a one cross state, and a reverse cross state. The seat device **10** can convert the state of the seat **1** between the long state (refer to FIG. **9** and FIG. **6**) in which the back of the seat is substantially parallel to and along a wall A, and a cross state (refer to FIG. **5**) in which the back of the seat is

substantially orthogonal to the wall A. Here, for the cross state, there are one cross state (refer to FIG. **5**), and a reverse cross state that is 180 degrees in the opposite direction to the one cross state.
(49) As shown in FIG. **15**, when it is assumed that the long state of the seat **1** has a rotation angle of 0 degrees as an original position, the rotation angle of the one cross state is 90 degrees, and the rotation angle of the reverse cross state is −90 degrees. Note that the back of the seat is synonymous with the back of the backrest **3**. Hereinafter, when collectively referring to the cross state and the reverse cross state, they are merely written as the cross state.
(50) <Leg Stand **11**>
(51) As shown in FIG. **10**, the leg stand **11** is fixed onto the floor surface near the wall A in the cabin. The leg stand **11** is formed by combining frame members into the shape of a stand that is long in the direction (front and back direction) substantially orthogonal to the wall A. Although an upper surface side of the leg stand **11** is substantially horizontal, and this upper surface side is surrounded by both side ends **12** and **12**, forming the long sides, and a rear end portion, forming a short side on the rear side (wall A side), the front side (aisle side) is opened.
(52) The leg stand **11** is arranged so that its rear end portion is close to and substantially parallel to the wall A, and both side ends **12** and **12** are substantially orthogonal to the wall A and extend toward the aisle side. Note that, in addition to the sliding mechanism **14**, which will be described next, related parts such as a stopper for regulating the advance and retract range and the rotation direction of the underframe **30** are provided in the upper surface side of the leg stand **11**.
(53) <Sliding Mechanism **14**>
(54) As shown in FIG. **10**, the movable stand **20** is attached to the upper surface side of the leg stand **11** via the sliding mechanism **14**, so as to be able to advance and retract in the direction substantially orthogonal to the wall A. The sliding mechanism **14** includes a pair of guide rails **14***a* and **14***a* that are provided inside both side ends **12** and **12** of the leg stand **11**. A pair of guide rails **14***a* and **14***a* are parallel to and oppose to each other along the inside of both side ends **12** and **12** of the leg stand **11**, and both side portions **21** and **21** of the movable stand **20**, which will be described next, directly and slidably fit inside the respective guide rails **14***a*.
(55) <Movable Stand **20**>
(56) As shown in FIG. **10**, the movable stand **20** is substantially horizontally arranged on the upper surface side of the leg stand **11**, and is formed by combining frame members into a rectangular framework shape. Both side ends **21** and **21**, forming the long sides of the movable stand **20**, slidably fit inside the aforementioned pair of guide rails **14***a* and **14***a*. Thus, the movable stand **20** can be slid so as to advance or retract in the direction substantially orthogonal to the wall A. The rotation mechanism **40** that rotates the seat **1** about the rotation axis is provided at a substantially center of the movable stand **20**.
(57) <Rotation Mechanism **40**>
(58) The rotation mechanism **40** supports the underframe **30** of the seat **1** on the movable stand **20** so as to be rotatable in the forward and backward directions in a substantially horizontal plane. The rotation mechanism **40** is formed as a unit in which, for example, a pair of inner and outer ring-shaped turntables are rotatably combined with each other by interposing a bearing, etc. between them. In this rotation mechanism **40**, the outer turntable is fixed to the moving table **20**, and the inner turntable is fixed to the underframe **30**.
(59) The rotation axis, which serves as the rotation center of the seat **1**, is the center line of the rotation mechanism **40**, and does not have a physical substance in the present embodiment. As shown in FIG. **8**, the rotation mechanism **40** includes a motor **41**, which is a power source. The motor **41** is provided with a reducer, and a drive gear in its output axis is rotatably engaged with a sprocket **42** centered on the rotation axis provided in the underframe **30** side. Note that the rotation mechanism **40** also allows the seat to be manually rotated.
(60) <Underframe **30**>
(61) As shown in FIG. **10**, the seat **1** is attached to the underframe **30**, and the underframe **30** is

supported by the rotation mechanism **40** on the movable stand **20**. The underframe **30** is formed by, for example, a metal plate that corresponds to a bottom surface of the seating portion **2**. Although described above, the sprocket **42** with which the drive gear of the motor **41** rotatably engages is integrally provided in the bottom surface side of the underframe **30**.
(62) <Interlocking Mechanism>
(63) Additionally, the seat device **10** includes an interlocking mechanism (not shown) that interlocks the rotation and advancement and retraction of the seat **1**, so that the seat **1** does not interfere with the wall A, when converting the seat **1** to the long state, the one cross state, and the reverse cross state. Note that the long state corresponds to "the specific state of the seat **1**" of the present invention.
(64) When the seat **1** is rotated with the underframe **30**, the interlocking mechanism converts the rotation of the underframe into linear motion, transmits the linear motion to the movable stand **20**, and makes the movable stand **20** move in a linear direction so as to be close to or separated from the wall A together with the underframe **30**. Although the kind of such an interlocking mechanism is not particularly limited, specifically, for example, the invention already proposed by the present applicant and described in Japanese Patent Laid-Open No. 2018-187971 may be utilized, or, although not published, the invention proposed in Japanese Patent Application No. 2019-239066, etc. may be utilized.
(65) <Rotation Lock Mechanism **60**>
(66) FIG. **13** is a perspective view showing a rotation lock mechanism **60**. FIG. **14** is a perspective view showing a state where the rotation lock mechanism **60** is seen from the bottom.
(67) As shown in FIG. **10**, the seat device **10** includes a rotation lock mechanism **60** that unrotatably restrains the underframe **30** (seat) in each rotation position of the long state, the one cross state, and the reverse cross state. Since the rotation lock mechanism **60** unrotatably locks the underframe to the leg stand **11**, the movable stand **20** is also inevitably restrained to the leg stand **11** so as not to be able to advance and retract.
(68) The rotation lock mechanism **60** includes a lock pin **61** that can protrude up and down from the leg stand **11** side to the underframe **30**, and locking holes **62***a*, **62***b*, and **62***c* that are provided in the underframe **30**, and with and from which the lock pin **61** are engaged and released. A total of three locking holes **62***a*, **62***b*, and **62***c* are provided in a long side along the back of the seat, and both short sides along the seat of the substantially rectangular underframes **30**, respectively.
(69) The lock pin **61** is incorporated in a unit **60***a*, and the unit is fixed near the rear end of the upper surface side of the leg stand **11**. The lock pin **61** is operated between a lock position at which the lock pin **61** can protrude upward from the upper surface side of the leg stand **11** to project upward and fit into the locking holes **62***a*, **62***b*, and **62***c*, and a lock release position at which the lock pin **61** retracts downward to be released from the locking holes **62***a*, **62***b*, and **62***c*.
(70) When the seat **1** is converted into the long state, the one cross state, and the reverse cross state, the lock pin **61** unrotatably restrains the seat **1** by fitting into the locking holes **62***a*, **62***b*, and **62***c* on the underframe **30** side to which the lock pin **61** vertically corresponds at the respective positions. That is, in the long state, the lock pin **61** fits into the locking hole **62***a* in one long side of the underframe **30**. Additionally, in the one cross state, the lock pin **61** is inserted into and engaged with the locking hole **62***b* in one short side of the underframe **30**. Furthermore, in the reverse cross state, the lock pin **61** fits into the locking hole **62***c* in the other short side of the underframe **30**.
(71) As shown in FIG. **13** and FIG. **14**, the unit **60***a* in which the lock pin **61** is incorporated is provided with each of a spring member (not shown) that always urges the lock pin **61** to project upward to the lock position, and a link **60***b* that makes the lock pin **61** resist the biasing force of the spring member to retract to the downward lock release position. Here, one end side of a rotation operation cable **206** (see FIG. **11**) for step operation is connected to the link **60***b*.
(72) The lock pin **61** is configured to be normally maintained at a locking position by the biasing force of the spring member, and to retract to a lock release position against the biasing force of the

spring member, when the link **60***b* is pulled by the rotation operation cable **206**. As shown in FIG. **11**, the other end side of the rotation operation cable **206** extends to the rotation operation unit **200** provided in the leg stand **11** side, which will be described later. Here, when the rotation operation cable **206** is pulled by the operation in the rotation operation unit **200**, the restraint of the rotation lock mechanism is released.

(73) Additionally, although illustration is omitted, one end side of a rotation operation cable for electric operation is also connected to the link **60***b* of the rotation lock mechanism **60**. For example, the motor **41** of the rotation mechanism **40** also serves as the power source for pulling the rotation operation cable for electric operation. That is, the motor **41** includes a clutch, and is configured to be able to switch between an operation for rotating the seat by the rotation mechanism **40**, and an operation for retracting the lock pin **61** to release the lock, by switching of the clutch. Note that a detailed description of the clutch of the motor **41** is omitted, since the configuration regarding the clutch of the motor **41** is common.

(74) The rotation lock mechanism **60** according to the present embodiment is configured such that, when the seat **1** is in the long state, the restraint by the rotation lock mechanism **60** cannot be released by the step operation in the rotation operation unit **200**, and can be released only by an electric operation by the motor **41**. Here, the electric operation is performed by a crew or station employee of a vehicle, and the step operation is mainly performed by a passenger.

(75) <Rotation Operation Unit **200**>

(76) FIG. **11** is a side view showing a state where the rotation operation unit **200** is not restrained by the rotation operation prevention mechanism **210** to disable the operation. FIG. **12** is a side view showing a state where the rotation operation unit **200** is restrained by the rotation operation prevention mechanism **210** to disable the operation.

(77) As shown in FIG. **5**, the rotation operation unit **200** is unitized by incorporating each part in one housing **201**, and is fixed to hang from a front end side of the movable stand **20**.

(78) As shown in FIG. **11**, the rotation operation unit **200** includes a step pedal **204** swingably supported by a lower end of a supporting bracket **202** fixed to the inside of the housing **201** via an axis **203**. A tip side of the step pedal **204** can be swung in the front and back directions, with the axis **203**, to which its base end side is pivotably supported, being the rotation center.

(79) The step pedal **204** is swung between a using position (refer to FIG. **11**) at which the tip side protrudes forward of the housing **201**, and a housing position (refer to FIG. **12**) at which the tip side withdraws upward. The step pedal **204** is normally urged to protrude forward to be in the using position via the spring member **205**. Here, when the step pedal **204** is in the using position, an operation of stepping on this to release the restraint by the rotation lock mechanism **60** is enabled, but when the step pedal **204** is in the housing position, the operation of releasing the restraint by the rotation lock mechanism **60** is disabled.

(80) The base end side of the step pedal **204** is connected to the other end side of the rotation operation cable **206** for step operation extended to the rotation lock mechanism **60** side via a connector. Here, when the step pedal **204** in the using position is stepped downward, the rotation operation cable **206** is pulled and the lock pin **61** (refer to FIG. **13**) is retracted downward, and the locked state of the rotation lock mechanism **60** is released.

(81) Additionally, a pin-shaped engaged portion **207** projecting in both directions at a position eccentric from the axis **203** is fixed to the base end side of the step pedal **204**. The engaged portion **207** is connected to the other end side of the reclining lock cable **124** extending from the reclining operation lock mechanism **120** side via a connector. Here, when the engaged portion **207** is engaged with an engaging portion **211**, which will be described next, the reclining lock cable **124** is pulled, the stopper **121** of the reclining operation lock mechanism **120** is moved backward, and the reclining operation unit **100** is restrained to disable the operation.

(82) <Rotation Operation Prevention mechanism **210**>

(83) Additionally, the rotation operation unit **200** is provided with a rotation operation prevention

mechanism **210** that disables the release operation of the restraint by the rotation operation unit **200**. The rotation operation prevention mechanism **210** disables the operation of the step pedal **204** when the seat **1** is in the long state. The long state here corresponds to a “the seat **1** is in a specific state” in the present invention.
(84) As shown in FIG. **11** and FIG. **12**, the rotation operation prevention mechanism **210** includes the engaging portion **211** provided in the leg stand **11**, which is the fixed side of the seat **1**, and the engaged portion **207** provided in the rotation operation unit **200** in the movable side of the seat **1**. The engaging portion **211** is arranged in the front end of the bottom surface side of the leg stand **11**, and is formed in, for example, a metal bracket shape that protrudes diagonally upward toward the front.
(85) The engaged portion **207** is provided in the metal pin shape at the position eccentric from the axis **203** in the base end side of the step pedal **204** as described above. The engaged portion **207** is set to precisely engage with the engaging portion **211** when the seat **1** in the long state (specific state). Since the step pedal **204** is swung to be in the housing position against the biasing force of the spring member **205** when the engaged portion **207** is engaged with the engaging portion **211**, the release operation in the rotation operation unit **200** is disabled. At the same time, the reclining operation unit **100** is restrained to disable the operation.
(86) <Effects of Seat Device **10**>
(87) Hereinafter, effects of the seat device **10** according to the present embodiment will be described.
(88) First, based on FIG. **15**, the operation of converting the state of the seat **1** will be described. As shown in FIG. **15** (a), when the underframe **30** (seat **1**) is in the long state, the rotation axis (rotation mechanism **40**) of the underframe **30** is most retracted (close) to the wall A side. The long side of the underframe **30** (the back of the seat) is substantially parallel to and along the wall A, and the rotation angle is 0 degrees. At this time, as shown in FIG. **10**, the lock pin **61** of the rotation lock mechanism **60** fits into the locking hole **62***a* in one long side of the underframe **30**, and the underframe **30** (seat **1**) is unrotatably restrained in the long state.
(89) «Conversion from Long State to One Cross State»
(90) As shown in FIG. **15**(*a*) to FIG. **15**(*c*), in order to convert the underframe **30** (seat **1**) from the long state into the one cross state (the rotation angle 90 degrees), it is necessary to release the restraint of rotation by the rotation lock mechanism **60**. The operation of disengaging the lock pin **61** from the locking hole **62***a* cannot be performed by an operation of the step pedal **204**, and is performed by an electric operation utilizing the power of the motor **41**.
(91) In the long state shown in FIG. **15** (*a*), when the underframe **30** is rotated by the motor **41** to the forward direction (the counter clockwise direction in FIG. **15**) as shown in FIG. **15** (*b*) after releasing the restraint by the rotation lock mechanism **60**, the underframe **30** is rotated while moving forward by interlocking mechanism. That is, the underframe **30** is rotated while being rotated in the forward direction and moving forward to the aisle side, so as not to interfere with the wall A.
(92) As shown in FIG. **15**(*c*), when the underframe **30** reaches the one cross state (the rotation angle 90 degrees), in FIG. **10**, the lock pin **61** of the rotation lock mechanism **60** fits into the locking hole **62***b* in one short side of the underframe **30**. Accordingly, the underframe **30** (seat **1**) is unrotatably restrained in the one cross state.
(93) «Conversion from One Cross State to Reverse Cross State»
(94) In the one cross state shown in FIG. **15**(*c*), when the underframe **30** is rotated to the opposite direction (the clockwise direction in FIG. **15**) after releasing the restraint of the rotation lock mechanism **60**, the underframe **30** advances or retracts while, for example, being rotated by the interlocking mechanism. Subsequently, as shown in FIG. **15**(*d*), in a state where the underframe **30** is temporarily held at the position to which the underframe **30** has advanced in a state parallel to the long state, the underframe **30** is directly rotated to the opposite direction without advancing and

retracting.
(95) As shown in FIG. **15** (*e*), when the underframe **30** reaches the reverse cross state (the rotation angle −90 degrees), the lock pin **61** of the rotation lock mechanism **60** fits into the locking hole **62***c* in the other short side of the underframe **30**, and the underframe **30** is restrained again to disable the rotation. Such conversion from the one cross state to the reverse cross state of the seat **1** can be performed not only by the step operation of the step pedal, but also by the electric operation of the motor **41**. Note that, in order to return the seat **1** from the reverse cross state to the one cross state, and further from the one cross state to the original long state, the inverse operations of the aforementioned conversion from the long state to the one cross state, and from the one cross state to the reverse cross state may be performed, respectively.
(96) «Restraint of Reclining Operation of Backrest **3**»
(97) When the seat **1** is in the long state, the reclining operation in the reclining operation unit **100** is disabled by the reclining operation lock mechanism **120**. That is, when the seat **1** is converted into the long state, as shown in FIG. **12**, the engaged portion **207** is engaged with the engaging portion **211** in the leg stand **11** to be displaced. Here, the engaged portion **207** is displaced in a direction that causes the step pedal **204** to swing upward by using the axis **203** as a rotation center. With displacement of the engaged portion **207**, the reclining lock cable **124** connected to the engaged portion **207** is pulled.
(98) In FIG. **1**(*a*), when the reclining lock cable **124** is pulled, the stopper **121** is linearly moved from the unrestrained position at which the stopper **121** is released from the abutting portion **113** of the operation lever **110** to the restrained position at which the stopper **121** is engaged with the abutting portion **113** against the biasing force of the spring member **123** as shown in FIG. **1**(*b*). Accordingly, the operation lever **110** remains at the ordinary initial position, cannot cause the operation unit **112** to swing upward, is restrained to disable the operation, and cannot tilt the backrest **3** by the reclining mechanism **50**.
(99) With the reclining operation lock mechanism **120** as described above, in the long state, it is possible to reliably prevent the reclining operation of the backrest **3** by a seated person, and to prevent the backrest **3** from being carelessly tilted to interfere with the wall A. The reclining operation lock mechanism **120** may only have the stopper **121** that is directly pulled by the reclining lock cable **124**, and it becomes unnecessary to have a special mechanism, such as a link for linearly moving the stopper **121**.
(100) Accordingly, the number of parts of the reclining operation lock mechanism **120** is reduced so that the configuration is also simplified and downsized, and it becomes possible to reduce the cost and the arrangement space. Additionally, the stopper **121** is not indirectly pulled by the reclining lock cable **124** via another member such as a link, but is directly connected to and is directly pulled by the reclining lock cable **124**. Therefore, there is no risk that pulling becomes insufficient due to deterioration of members other than the reclining lock cable **124**, and the stopper **121** can be reliably moved.
(101) Moreover, the stopper **121** is linearly moved in the front and back directions on the substantially horizontal bottom wall **103** of the housing **101**. Additionally, the reclining lock cable **124** directly connected to the stopper **121** also linearly extends in the same direction on the same horizontal plane on which the stopper **121** is linearly moved. Accordingly, the reclining operation lock mechanism **120** does not have a configuration and an operation that are bulky in the vertical direction, and can significantly suppress the height dimension. Therefore, it becomes possible to add, on the upper end side of the sleeve portion **4**, the armrest **5** having a thickness that corresponds to the amount of reduction of the size of the reclining operation lock mechanism **120** in the vertical direction.
(102) «Release of Restriction of Reclining Operation of Backrest **3**»
(103) When the seat **1** is converted into the cross state, the restraint of the reclining operation unit **100** by the reclining operation lock mechanism **120** to disable the operation is released. That is, in

the cross state of the seat **1**, as shown in FIG. **11**, the engaged portion **207** of the step pedal **204** in the movable stand **20** is not engaged with and is separated from the engaging portion **211** in the leg stand **11**. Thus, the step pedal **204** protrudes forward to be in the using position by the biasing force of the spring member **205**, and the reclining lock cable **124** is in a state where the reclining lock cable **124** is not pulled.
(104) Then, as shown in FIG. **1**(*a*), in the reclining operation lock mechanism **120**, the stopper **121** is linearly moved forward to the unrestrained position at which the stopper **121** is released from the abutting portion **113** of the operation lever **110** by the biasing force of the spring member **123**. Accordingly, the restraint of the operation lever **110** to disable the operation is released, and the reclining operation is enabled. That is, as shown in FIG. **4**, a seated person can perform the reclining operation that causes the operation unit **112** of the operation lever **110** to swing upward, can tilt the backrest **3** to an arbitrary angle, and can improve seating comfort.
(105) Note that, when the operation unit **112** of the operation lever **110** is swung to the upper operation position, the middle of the reclining lock cable **124** that penetrates through the concave groove **114** of the abutting portion **113** is pressed downward. This displacement of the reclining lock cable **124** is set to be within a range that is absorbed by play of the reclining lock cable **124**. Accordingly, the operation of the rotation operation unit **200** that is located ahead of the reclining lock cable **124** is not affected.
(106) «Restraint of Rotation Operation of Seat **1**»
(107) As shown in FIG. **10**, when the seat **1** is in the long state, the underframe **30** is restrained to disable the rotation with respect to the movable stand **20** by the rotation lock mechanism **60**. That is, the lock pin **61** of the rotation lock mechanism **60** fits into the locking hole **62***a* in the one long side of the underframe **30**. Here, since the lock pin **61** protrudes from the leg stand **11**, which is the fixed side of the seat **1**, the seat **1** is not only restrained to disable the rotation, but also simultaneously restrained to disable advancement and retraction.
(108) As shown in FIG. **6**, when the seat **1** is in the long state, the release operation in the rotation operation unit **200** is disabled by the rotation operation prevention mechanism **210**. That is, in the long state, the movable stand **20** is most retracted (close) to the wall A side, and the front end side of the movable stand **20** overlaps with the front end side of the leg stand **11**. With this positional relationship, as shown in FIG. **12**, the engaged portion **207** of the step pedal **204** in the movable stand **20** is engaged with the engaging portion **211** in the leg stand **11**. Then, the step pedal **204** is swung to be in the housing position at which the step pedal **204** is upright against the biasing force of the spring member **205**, and is restrained in the housing position.
(109) Accordingly, when the seat **1** is in the long state, the step pedal **204** is not only displaced to the housing position at which the operation is disabled, but also firmly held in the housing position by the engagement relationship between the engaging portion **211** and the engaged portion **207**. Thus, in the long state, the release operation in the rotation operation unit **200** is disabled. With such a simple configuration, the rotation operation of the seat **1** by a seated person can be reliably prevented in the long state.
(110) «Release of Restriction of Rotation Operation of Seat **1**»
(111) When the seat **1** is in the cross state, the restraint to disable the operation of the rotation operation unit **200** by the rotation operation prevention mechanism **210** is also released. That is, in the cross state shown in FIG. **15** (*c*), the movable stand **20** is most advanced (separated) from the wall A side, and the front end side of the movable stand **20** is located farther forward from the front end side of the leg stand **11**. In this positional relationship, as shown in FIG. **11**, the engaged portion **207** of the step pedal **204** in the movable stand **20** is not engaged with and separated from the engaging portion **211** in the leg stand **11**. Thus, the step pedal **204** protrudes forward to be in the using position by the biasing force of the spring member **205**.
(112) At this time, the seated person of the seat **1** can release the restraint by the rotation lock mechanism **60** by stepping on the step pedal **204**. That is, in FIG. **11**, when the step pedal **204** is

stepped on to be swung downward, the rotation operation cable **206** is pulled, the lock pin **61** (refer to FIG. **13**) is retracted downward, and the locked state of the rotation lock mechanism **60** is released. Accordingly, the seated person can manually rotate the seat **1**.
(113) <Configuration and Effects of Present Invention>
(114) Although the embodiment of the present invention has been described above, the present invention is not limited to the aforementioned embodiment. The present invention derived from the aforementioned embodiment will be described below.
(115) First, the present invention is the seat device **10** that can convert the state of the seat **1**, including the reclining mechanism **50** that can tilt the backrest **3** of the seat **1**, the operation unit **100** that performs the operation of tilting the backrest **3** by the reclining mechanism **50**, the lock mechanism **120** that restrains the operation unit **100** to disable the operation when the seat **1** is in the specific state, and the cable **124** that is pulled in conjunction with the conversion into the specific state of the seat **1**, wherein the operation unit **100** can be displaced from the ordinary initial position to the operation position at a time of the operation, the lock mechanism **120** includes the stopper **121** that is linearly moved to be engaged with and released from the operation unit **100** in the initial position, and the stopper **121** is linearly moved from the ordinary unrestrained position at which the stopper **121** is directly pulled by the cable **124** to be released from the operation unit **100** to the restrained position at which the stopper **121** is engaged with the operation unit **100** to disable the operation.
(116) In the lock mechanism **120** as described above, the stopper **121** is directly pulled by the cable **124**, and is linearly moved to the restrained position at which the stopper **121** is engaged with the operation unit **100** of the reclining mechanism **50** to disable the operation. Therefore, a special mechanism, such as a link, for moving the stopper **121** is no longer required, the number of parts is reduced so that the configuration is also simplified and downsized, and it becomes possible to reduce the cost and the arrangement space.
(117) Moreover, the stopper **121** is not indirectly pulled by the cable **124** via another member such as a link, but is directly connected to and is directly pulled by the cable **124**. Therefore, there is no risk that pulling becomes insufficient due to deterioration of members other than the cable **124**, and the stopper **121** can be reliably moved.
(118) Additionally, as the present invention, the stopper **121** is supported in a state where the stopper **121** is linearly moved in the front and back directions of the seat **1** on a substantially horizontal plane.
(119) Accordingly, the reclining operation lock mechanism **120** does not have a configuration and an operation that are bulky in the vertical direction, and can significantly suppress the height dimension. Accordingly, it is possible to further respond to a request of space-saving in the arrangement space of, for example, the sleeve portion **4** (armrest **5**) or the like to which the reclining operation lock mechanism **120** is attached, and the design freedom can also be increased.
(120) Additionally, as the present invention, the stopper **121** is normally held at the unrestrained position by the biasing means **123**, and when the stopper **121** is pulled by the cable **124**, the stopper **121** is linearly moved to the restrained position against the biasing force of the biasing means **123**, and is held at the restrained position.
(121) By holding the stopper **121** at the unrestrained position by the biasing force of the biasing means **123** in this manner, it is possible to easily maintain the stopper **121** at the unrestrained position with a simple configuration, without using power. Additionally, it is possible to reliably hold the stopper **121** at the restrained position by easily and linearly moving the stopper **121** to the restrained position against the biasing force of the biasing means **123**, only by pulling the cable **124**.
(122) Additionally, as the present invention, the operation unit **100** includes the operation lever **110** that is swingably supported between the initial position and the operation position at the front end side of the armrest **5** in the seat **1**, the operation lever **110** is provided with the abutting portion **113**

that is engaged with and released from the stopper **121**, the abutting portion **113** being provided on an other end side of the operation lever **110**, which is on the other side of the oscillation center with respect to the one end side of the operation lever **110** to be operated, and the cable **124** is routed in the state where a straight portion of the one end side of the cable **124** penetrates through the abutting portion **113** of the operation lever **110**, and is directly connected to the stopper **121** arranged ahead of the abutting portion **113**.
(123) Since the operation unit **100** of the reclining mechanism **50** includes the operation lever **110** on the front end side of the armrest **5** in the seat **1** in this manner, a seated person can easily operate the operation lever **110** while remaining seated. The one end side of the operation lever **110** serves as the portion (operation unit **112**) to be operated, and the other end side serves as the abutting portion **113** to be engaged with and released from the stopper **121**, with the oscillation center being between the one end side and the other end side. The cable **124** is routed in the state where the cable **124** penetrates through the abutting portion **113** (concave groove **114**) of the operation lever **110**, and is directly connected to the stopper **121** located ahead of the abutting portion **113**.
(124) With this configuration, the stopper **121** is arranged between the one end side and the other end side of the operation lever **110**, and the cable **124** is arranged so as to overlap with the other end side of the operation lever **110**. Therefore, it becomes possible to generally more compactly form the operation unit **100** and the lock mechanism **120**. Here, it also becomes possible to reliably move the stopper **121** with a small force by matching the direction in which the stopper **121** is pulled and the direction in which the stopper **121** is linearly moved.
(125) Additionally, the present invention includes the rotation mechanism **40** that rotates the seat **1** about the rotation axis, and the sliding mechanism **14** that causes the seat **1** to advance and retract from the fixed side, together with the rotation mechanism **40**, wherein by interlocking of the rotation mechanism **40** and the sliding mechanism **14**, the state of the seat **1** can be converted into a long state where a back of the seat **1** is substantially parallel to and along the wall, and a cross state where the back of the seat **1** is separated from the wall in the orientation substantially orthogonal to the long state, and the specific state of the seat **1** is the long state.
(126) Accordingly, as described in the aforementioned embodiment, it becomes possible to directly apply the present seat device **10** to general stool-type rotary seats mounted in railroad cars. Then, since the back of the backrest **3** is close to and substantially parallel to the wall A when the seat **1** is in the long state, as described above, it is necessary to restrain the operation unit **100** of the reclining mechanism **50** to disable the operation by the lock mechanism **120**.
(127) Further, the present invention includes the leg stand **11** that is the fixed side of the seat **1**, the movable stand **20** supported by the leg stand **11** so as to be able to advance and retract via the sliding mechanism **14**, the seat **1** being rotatably supported by the movable stand **20** via the rotation mechanism **40**, the engaging portion **211** provided in the leg stand **11**, and the engaged portion **207** provided in the movable stand **20**, the engaged portion **207** being engaged with the engaging portion **211** to be displaced when the seat **1** is in the long state, wherein the engaged portion **207** and the stopper **121** are connected by the cable **124**, and the cable **124** is pulled based on displacement of the engaged portion **207**.
(128) In this manner, according to the present seat device **10**, it is possible to move the stopper **121** via the cable **124**, due to the mechanical engagement relationship between the engaging portion **211** and the engaged portion **207**. With such a simple configuration, the reclining operation can be regulated only in the long state, which is the specific state of the seat **1**, without using electric power.
(129) <Another Configuration and Effects of Present Invention>
(130) Additionally, the following another invention is also derived from the aforementioned embodiment. the seat device **10** that can convert the state of the seat **1**, including the rotation lock mechanism **60** that can restrain the seat **1** at each of a plurality of rotation angles, the seat being rotatable about the rotation axis, the rotation operation unit **200** that performs the release operation

of restraint by the rotation lock mechanism **60**, and the rotation operation prevention mechanism **210** that disables the release operation in the rotation operation unit **200**, when the seat **1** is in the specific state of one of the rotation angles, the rotation operation prevention mechanism **210** including the engaging unit **211** provided on the fixed side of the seat **1**, and the engaged unit **207** that is provided in the rotation operation unit **200** on the movable side of the seat **1**, and disables the release operation in the rotation operation unit **200** by being engaged with the engaging unit **211** when the seat **1** is in the specific state.
(131) In the present seat device **10**, when the seat **1** is in the long state, which is the specific state, although the back of the seat is close to the wall A, the rotation of the seat **1** by a seated person is disabled, so that a passenger cannot freely change the position. Here, it is necessary to release the restraint by the rotation lock mechanism **60** by the rotation operation unit **200** for rotating the seat **1**. Therefore, the rotation of the seat **1** can be disabled by disabling the operation of the rotation operation unit **200** by the rotation operation prevention mechanism **210**.
(132) In the rotation operation prevention mechanism **210**, when the seat **1** is in the long state, the release operation of the rotation operation unit **200** is disabled by engaging the engaged portion **207** in the rotation operation unit **200** on the movable side of the seat **1** with the engaging portion **211** in the fixed side of the seat **1**. With such a mechanical engagement relationship between the engaging portion **211** and the engaged portion **207**, the rotation of the seat **1** by a seated person can be disabled only when the seat **1** is in the long state, with a simple configuration, and without using electric power.
(133) Although the embodiments have been described above with the drawings, the specific configuration is not limited to these embodiments, and even when there are modification and addition in the scope not departing from the gist of the present invention, they are included in the present invention. For example, although the example of the seat **1** for two persons has been described, the seat **1** may be for three persons or one person.
(134) Additionally, the specific shapes of the operation lever **110** of the reclining operation unit **100** and the stopper **121** of the reclining operation lock mechanism **120** are not limited to those shown. In addition, the specific shapes of the leg stand **11**, the movable stand **20**, and the underframe **30** are not limited to those shown, either. Furthermore, the conversion of the state of the seat is not limited to the long state, the one cross state, and the reverse cross state.
INDUSTRIAL APPLICABILITY
(135) The present invention can be widely utilized as a seat device for chairs for theaters, home, and office, in addition to the seat for vehicles installed in cabins of railroad cars, airplanes, automobiles, marine vessels, etc.
REFERENCE SIGNS LIST
(136) **10** . . . device **11** . . . Leg stand **14** . . . Sliding mechanism **20** . . . Movable stand **30** . . . Underframe **40** . . . Rotation mechanism **50** . . . Reclining mechanism **60** . . . Rotation lock mechanism **100** . . . Reclining operation unit **110** . . . Operation lever **111** . . . Axis **112** . . . Operation unit **113** . . . Abutting portion **116** . . . Reclining operation cable **120** . . . Reclining operation lock mechanism **121** . . . Stopper **122** . . . Axis **123** . . . Spring member **124** . . . Reclining lock cable **200** . . . Rotation operation unit **204** . . . Step pedal **207** . . . Engaged portion **210** . . . Rotation operation prevention mechanism **211** . . . Engaging portion

Claims

1. A seat device that can convert a state of a seat disposed in a vehicle, comprising: a reclining mechanism that can tilt a backrest of the seat; an operation unit that enables or disables an operation of the reclining mechanism to tilt the backseat; a lock mechanism that restrains the operation unit to disable the operation when the seat is arranged in a specific state relative to the vehicle; and a cable that is pulled when the seat is arranged in the specific state, wherein the

operation unit can be displaced from an initial position to an operation position at a time of the operation, the lock mechanism includes a stopper that is linearly moved to be engaged with and released from the operation unit in the initial position, and the stopper is connected to the cable to be linearly moved from an ordinary unrestrained position at which the operation unit is unrestrained from the lock mechanism to enable the operation to a restrained position at which the operation unit is restrained by the lock mechanism to disable the operation.

2. The seat device according to claim 1, wherein the stopper is supported in a state where the stopper is linearly moved in front and back directions of the seat on a substantially horizontal plane.

3. The seat device according to claim 1, wherein the stopper is normally held at the unrestrained position by biasing means, and when the stopper is pulled by the cable, the stopper is linearly moved to the restrained position against a biasing force of the biasing means, and is held at the restrained position.

4. The seat device according to claim 1, wherein the operation unit includes an operation lever that is swingably supported between the initial position and the operation position at a front end side of an armrest in the seat, the operation lever is provided with an abutting portion that is engaged with and released from the stopper, the abutting portion being provided on an end side of the operation lever, and the cable penetrates through the abutting portion of the operation lever, and is directly connected to the stopper arranged ahead of the abutting portion.

5. The seat device according to claim 1, further comprising: a rotation mechanism that rotates the seat about a rotation axis; and a sliding mechanism that causes the seat to advance and retract from a fixed side, together with the rotation mechanism, wherein by interlocking of the rotation mechanism and the sliding mechanism, the state of the seat can be converted into a long state where the backrest is substantially parallel to and along a vertical side wall of the vehicle, and a cross state where the backrest is separated from the vertical side wall in an orientation substantially orthogonal to the long state, and the seat is the long state when the seat is arranged in the specific state.

6. The seat device according to claim 5, further comprising: a leg stand that is the fixed side of the seat; a movable stand supported by the leg stand so as to be able to advance and retract via the sliding mechanism, the seat being rotatably supported by the movable stand via the rotation mechanism; an engaging portion provided in the leg stand; and an engaged portion provided in the movable stand, the engaged portion being engaged with the engaging portion to be displaced when the seat is in the long state, wherein the engaged portion and the stopper are connected by the cable, and the cable is pulled based on displacement of the engaged portion.
